

57.pdf

 University of Engineering & Management

Document Details

Submission ID

trn:oid:::3618:103124122

Submission Date

Jul 1, 2025, 2:33 PM GMT+5:30

Download Date

Jul 5, 2025, 12:09 AM GMT+5:30

File Name

57.pdf

File Size

2.8 MB

12 Pages

3,425 Words

20,472 Characters

16% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

-  **25 Not Cited or Quoted 12%**
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations 0%**
Matches that are still very similar to source material
-  **8 Missing Citation 4%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 11%  Internet sources
- 13%  Publications
- 10%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- 25 Not Cited or Quoted 12%**
Matches with neither in-text citation nor quotation marks
- 0 Missing Quotations 0%**
Matches that are still very similar to source material
- 8 Missing Citation 4%**
Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 11% Internet sources
- 13% Publications
- 10% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Submitted works	Unizin, LLC on 2024-10-10	2%
2	Publication	Jan Jakub Kęsik, Wiesław Paja, Paweł Jakubczyk, Maryna Khalavka et al. "Determi...	1%
3	Publication	Bakr Ahmed Taha, S.A Abdulateef, Ali J. Addie, Suha. A. Muneam et al. "Paradigm ...	1%
4	Submitted works	University of Strathclyde on 2024-03-01	<1%
5	Submitted works	University of South Florida on 2023-05-05	<1%
6	Internet	www.photonics.com	<1%
7	Internet	www.hindawi.com	<1%
8	Publication	Katherine Boere, Francesca Anderson, Kent G. Hecker, Olav E. Krigolson. "Measur...	<1%
9	Internet	www.researchsquare.com	<1%
10	Internet	faculty.iitmandi.ac.in	<1%

11	Submitted works	University of Tennessee Knoxville on 2025-03-25	<1%
12	Publication	Dandan Zhao, Yancun Li, Keyi Ning, Bingjie Zou, Bin Wang, Libo Li, Qiaojun Zhang...	<1%
13	Publication	Symes, Stephen William Robert. "Simulator Based Human Performance Assessme...	<1%
14	Internet	repository.bsi.ac.id	<1%
15	Internet	www.medrxiv.org	<1%
16	Submitted works	Loughborough University on 2023-09-16	<1%
17	Publication	Ayan Chatterjee. "Sovereignty in Automated Stroke Prediction and Recommenda...	<1%
18	Publication	Lirui Yang, Zehua Wang. "Applications and advances of combined fMRI-fNIRs tec...	<1%
19	Internet	publications.eai.eu	<1%
20	Internet	ebin.pub	<1%
21	Internet	www.mdpi.com	<1%
22	Publication	"Cryptocurrency Market Investment Strategy and Analysis of Potential Opportuni...	<1%
23	Submitted works	University of Gloucestershire on 2025-05-30	<1%
24	Submitted works	CSU Northridge on 2021-03-19	<1%

25	Submitted works	Georgia Institute of Technology Main Campus on 2025-06-30	<1%
26	Internet	discovery.researcher.life	<1%
27	Internet	jscires.org	<1%
28	Publication	Mohamed Osman, Mohamed Ouf. "A comprehensive review of time use surveys i...	<1%
29	Publication	Yuning Cui, Weixuan Dong, Yifu Li, Amanda E. Janitz, Hanumantha R. Pokala, Rui ...	<1%
30	Publication	Jiachun xie, Dianhuan Tan, Tingting Zheng, Liya Wei. "Cardiac Echocardiographic ...	<1%

From Sensors to Survival: Mimicking a Non-Invasive System for Stroke Risk Stratification

Arshi Zameer^{1,2}, Sujata Pal¹, and Shubhajit Roy Chowdhury²

¹ Indian Institute of Technology Ropar, Punjab, 140001, India
arshi.23csz0011@iitrpr.ac.in, sujata@iitrpr.ac.in

² School of Computing and Electrical Engineering, Indian Institute of Technology
Mandi, Himachal Pradesh, 175005, India
src@iitmandi.ac.in

Abstract. Ischemic stroke is a significant health concern, especially in remote areas with limited access to advanced diagnostic equipment. This study presents a simplified, low-cost concept of a single-channel functional Near-Infrared Spectroscopy (fNIRS) system to monitor cerebral hemodynamic changes. Rather than building a complete physical device, the system behavior was mimicked through basic analog signal testing. In parallel, survival analysis was performed on a Kaggle stroke dataset to examine how clinical features influence stroke risk. Survival curves stratified risk groups effectively, with high-risk patients' survival probability dropping to 0.86 by day 250, unlike the low-risk group staying near 1.0. Cox model results highlighted age (Hazard Ratio ≈ 1.6) as the most significant predictor, followed by glucose level and hypertension. Violin plot showed median predicted risk ≈ 7 for stroke patients versus <2 for non-stroke cases. Correlation analysis revealed moderate links ($r \approx 0.28$) between age, hypertension, and heart disease.

Keywords: Functional near-infrared spectroscopy (fNIRS), stroke detection, ESP32, infrared LED, photodiode, signal acquisition, survival analysis

1 Introduction

Recent studies have stressed on the extensive health concern posed by Ischemic stroke, which accounts for almost 85-87% of all strokes and significantly contributes to the rising number of deaths and disabilities in developing countries like India [13]. Early detection of brain damage and its timely intervention are vital to improve recovery outcomes. MRI and CT scans are highly effective brain imaging techniques for stroke detection. However, as they are infrastructure dependent their use is limited in resource-constrained settings especially in rural and remote areas. As a result, Functional Near-Infrared Spectroscopy (fNIRS) has emerged as a promising alternative [10]. It offers a portable, cost-effective, and infrastructure-independent solution by non-invasively monitoring cerebral

2 Arshi Zameer et al.

blood oxygenation through changes in oxygenated and deoxygenated hemoglobin levels (Figure 1).

Commercial fNIRS systems are expensive and technically complex despite their potential [1] - [4], limiting their widespread adoption. Additionally, recent improvement in neural networks advanced techniques have improved fNIRS accuracy [14], strengthening its potential for real-time patient monitoring. This is particularly inspiring for point-of-care in remote areas. The potential of brain imaging with refined computational techniques is a thrilling prospect for the future of stroke risk monitoring innovations.

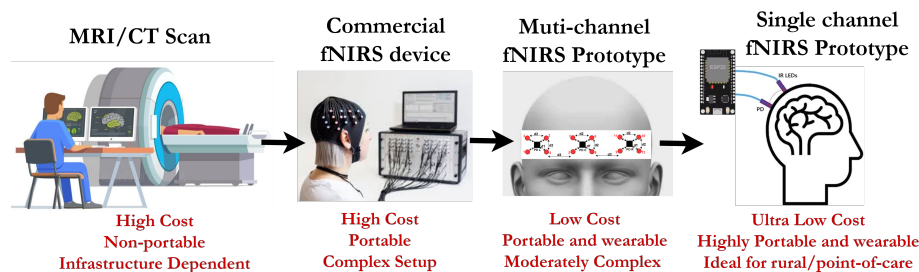


Figure 1: Evolution of brain monitoring techniques for stroke detection.

This study mimics a single-channel functional Near-Infrared Spectroscopy (fNIRS) system designed for stroke detection using a simplified hardware setup involving an ESP32, IR LED, and photodiode. The practical consequences of this research are important, as it offers a more affordable and easier way to detect strokes. Instead of full hardware implementation, the system behavior was conceptualized and represented through basic analog signal testing. To validate the study, survival analysis was conducted on a publicly available stroke prediction dataset to explore clinical risk patterns using Kaplan-Meier and Log-Rank methods.

2 Related Work

Recent research has explored the viability of low-cost and single-channel fNIRS configurations using minimal hardware to reduce system complexity and cost [12]. Wearable solutions like NeuroHAT [9] integrate fNIRS sensors with micro-controller platforms to achieve real-time monitoring yet still face limitations in accessibility and user-friendliness. Among the notable contributions in this area is the work by Ahirwar *et al.* [4], who developed a wearable, multichannel, multiwavelength NIRS system designed to capture hemodynamic biomarkers associated with ischemia. While Ahirwar's system demonstrated promising accuracy and validated its utility using in-vivo experiments and FDA-approved benchmarks, its multichannel configuration and layered probe design (which included

Non-Invasive System for Stroke Risk Stratification 3

sponge-rubber layering for probe mounting) introduced significant complexity, increased power demand, and higher costs making it less feasible for scalable deployment in rural and primary healthcare environments. Other researchers have also addressed similar goals of cost-effectiveness and portability. Xu et al. [6] validated a single-channel fNIRS system for cognitive monitoring tasks and found its performance comparable to commercial devices despite limitations in spatial resolution. Yang and Ge [12] similarly showed that a forehead-placed single-channel fNIRS signal could achieve up to 86% classification accuracy for BCI applications. These studies highlight the practicality of minimalist hardware while still capturing meaningful brain signal variations. Kumar et al. [8] introduced NIRduino, a Bluetooth-enabled, Arduino-based fNIRS system for low-cost applications. Their design emphasized modularity and mobile integration, although it lacked validation across diverse clinical scenarios. Liang et al. [9] extended this vision by developing NeuroHAT, a wearable system integrating EEG and fNIRS, validated using public and self-collected datasets. Though highly accurate, it remained dependent on multimodal input and lacked large-scale clinical validation. Beyond fNIRS-specific systems, Taha et al. [3] proposed a multiwavelength SCOS-optical spectroscopy system for high-resolution cerebral blood flow analysis, while Zhu et al. [5] combined Doppler and PPG sensors for arterial stenosis detection - both demonstrating strong physiological insights but at the cost of hardware complexity and simulation-based limitations.

Additionally, works like Liang et al. [1] and Cai et al. [7] reviewed the role of biological and neurophysiological biomarkers in ischemic stroke and chronic motor impairment, reinforcing the significance of hemodynamic signals in stroke diagnostics. Despite significant advances in fNIRS-based systems for brain monitoring and stroke detection, many existing approaches suffer from high hardware complexity, multi-sensor dependence, and limited applicability in low-resource or point-of-care environments. While some studies have explored low-cost alternatives, they often lack clinical validation or integration with predictive analytics. Table 1 presents a comparative analysis of prior systems, emphasizing the practical gap our work addresses - the need for a minimal, affordable, and analytically integrated fNIRS-inspired system for early stroke risk assessment.

The present study proposes a single-channel fNIRS prototype designed using the ESP32 microcontroller, infrared LEDs, and photodiodes - affordable and widely available components to overcome this. The system aims to detect hemodynamic changes associated with ischemic conditions, providing a practical solution for early stroke screening in low-resource settings. We strive to develop a simple, low-cost, clinically relevant platform that democratizes access to neuroimaging technologies. We drastically reduce component count, size, and power requirements by leveraging a combination of IR LEDs and a photodiode connected to an ESP32 microcontroller. This minimalist system still targets key hemodynamic features (ΔHbO_2 , ΔHb , ΔrSO_2) but with an architecture better suited to rural deployment, wearable integration, and low-cost manufacturing. In doing so, our contribution bridges the gap between clinical efficacy and practical accessibility - demonstrating that streamlined solutions can still retain diagnostic

4 Arshi Zameer et al.

Table 1: Comparison of prior fNIRS-based systems highlighting limitations that motivate the proposed low-cost, survival-driven approach

Study / System	Hardware Complexity	Channel Count	Cost & Portability	Validation Method	Key Limitations / Relevance
Taha et al. (2025)	Very High (Spectroscopy)	Multi-wavelength	Expensive, complex	Simulation only	High insight, but low deployability due to system complexity
Zhu et al. (2025)	High (PPG + Doppler)	Multi-sensor	Not minimal	Sensor combination testing	Complex setup, lacks point-of-care validation
Liang et al. (2024) – NeuroHAT	Very High (EEG + fNIRS)	Multi	Complex integration	Dataset-based validation	Accurate but not practical for rural clinical use
Kumar et al. (2024) – NIR-Duino	Medium	Multi	Mobile, modular	Signal transmission test	Modular, but lacks clinical validation
Xu et al. (2024)	Low (Single-channel)	Single	Affordable	Basic signal test	Limited resolution, but supports feasibility
Ahirwar et al. (2022)	High (Multi-channel, layered)	Multi	Expensive & complex	In-vivo + FDA benchmark	Accurate but not scalable for low-resource settings
This Work (Zameer et al.)	Low (Simulated)	Single	Minimal, ESP32-based	Signal mimicry + survival analysis	Real signals not yet acquired; enables low-cost early-stage validation

power. The paper is organized as follows: section first describes the introduction, followed by related work that summarizes the existing research and identifies the context of the current study. Section three presents the proposed methodology, detailing the research approach and methods used. Section four discusses the results obtained and we concluded by giving the direction toward the future scope.

3 Materials and Methods

3.1 Dataset Used

This study uses the *Stroke Prediction Dataset* from Kaggle, containing clinical features linked to stroke risk. The data was preprocessed for survival analysis using Kaplan-Meier [16] and Log-Rank tests. This dataset contains health-related records of individuals, including demographic, medical, and lifestyle information. Key attributes include age, gender, hypertension, heart disease, smoking

status, and average glucose levels. The target variable stroke indicates whether the individual has experienced a stroke (1: Yes, 0: No) [15].

3.2 Proposed Approach

As shown in Figure 2, our study comprehends a two-part approach to design and evaluate a basic one-channel functional Near-Infrared Spectroscopy (fNIRS) system for early stroke detection. The first part(A) focuses on building and simulating a prototype circuit to mimic fNIRS signal acquisition, and the second part(B) involves analyzing a real-world stroke dataset to identify relevant clinical patterns through survival analysis.

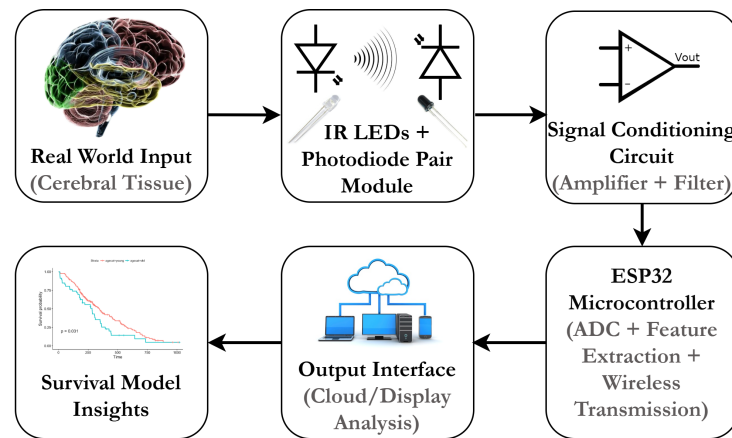


Figure 2: Block diagram of the proposed fNIRS system for stroke monitoring with ESP32 and survival model integration.

A. Circuit Simulation and Signal Acquisition: A current-regulated LED driver circuit, as shown in Figure 3 was used to mimic the signal acquisition mechanism of a basic fNIRS system. The setup employs an infrared LED for light emission and a photodiode for detection, configured in a reverse-biased arrangement. The constant current circuit ensures stable illumination intensity, critical for reliable optical measurements. As light passes through or reflects off the cerebral-inspired medium, the photodiode senses variations in light intensity corresponding to changes in tissue absorption. These variations are converted into voltage signals using a transimpedance stage.

The output from the photodiode circuitry is fed into the ESP32 microcontroller's analog-to-digital converter (ADC), which digitizes the signal for further processing. The ESP32 platform was selected for its integrated ADC, wireless capabilities, and ease of integration with custom sensor setups. This simulation

6 Arshi Zameer et al.

approach validates the ESP32's effectiveness in capturing and processing optically modulated signals, forming the foundation for our low-cost, portable fNIRS prototype.

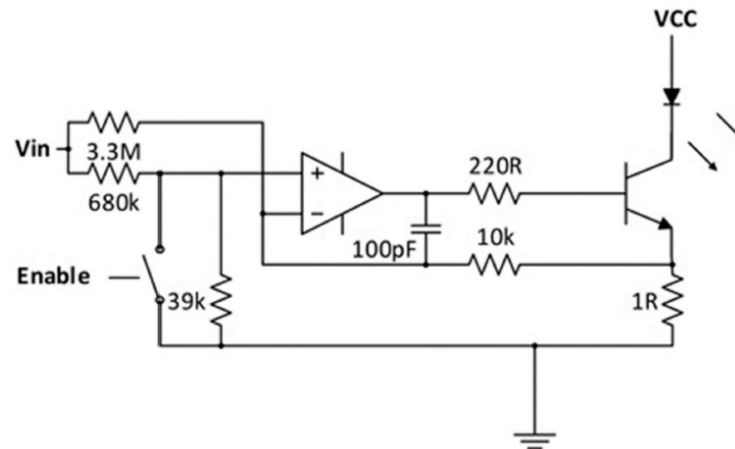


Figure 3: Simulated current regulator-based IR LED driver circuit with photodiode detection setup, forming the core of the proposed fNIRS signal acquisition module (Adapted from [11]).

B. Stroke Dataset Analysis Using Survival Methods: A survival analysis was performed using an openly available stroke dataset to support the hardware simulation with relevant clinical context. The goal was to study how certain risk factors influence stroke outcomes over time. Variables such as age, presence of heart disease, hypertension, and gender were used to group patients and evaluate their survival probabilities. The analysis was carried out in Python using the lifelines library. Kaplan-Meier survival curves were generated for different subgroups, and the log-rank test was used to estimate statistical differences. The results highlighted which clinical features had the most influence on stroke survival times, offering strong support for including such features in the design of early-warning systems. Overall, we propose a conceptual model for developing a wearable, point-of-care fNIRS device by integrating simulation insights and survival data interpretation. While real biological signal acquisition remains future work, this proof-of-concept lays a promising foundation for building low-cost early stroke risk assessment tools, especially in rural or resource-limited settings. The Python code implementing the survival analysis is available for public access on GitHub at <https://github.com/Erxi-17/fNIRS>.

Table 2 shows a thorough mapping between every hardware block and the matching code block that mimics its functionality after evaluating the conceptual hardware design.

Table 2: Mapping of Hardware Components to Code Implementation

Modules in Diagram	Mimicked in Code Block
Real World Input	<code>df = pd.read_csv(...)</code> – Simulated input from cerebral tissue (stroke dataset ingestion)
IR LEDs + Photodiode Module	<code>selected_cols = [...]</code> – Mimicking biosignal selection (e.g., glucose, age, etc.)
Signal Conditioning Circuit	<code>df[features] = (df - mean)/std</code> – Standardizing signals
ESP32 Microcontroller	Survival time simulation, event encoding, Cox model fitting and partial hazard prediction
Output Interface (Cloud/Display)	<code>df.to_csv(...)</code> , <code>risk_group = pd.qcut(...)</code> – Risk tiering and data export
Survival Model Insights	Correlation heatmap, Kaplan–Meier curves, violin plot, and Cox model coefficients

4 Results and Discussion

This section presents the results of the proposed dual-track methodology, combining a simplified signal acquisition model with survival analysis for stroke risk prediction. The discussion includes system feasibility, risk stratification, model performance, and clinical relevance.

4.1 Survival Analysis and Risk Evaluation

Numerous statistical techniques were applied to a publicly available stroke dataset for survival analysis. Kaplan–Meier survival curves (Figure 4) illustrated effective risk stratification. High-risk and medium-risk groups exhibited a significant drop in survival probability, showing around 87.5% by day 250 and 97.6% beyond 400 days, respectively. The low-risk group remained stable, with survival rates approaching 100%. These survival curves validate the model’s ability to classify patients by risk accurately.

8 Arshi Zameer et al.

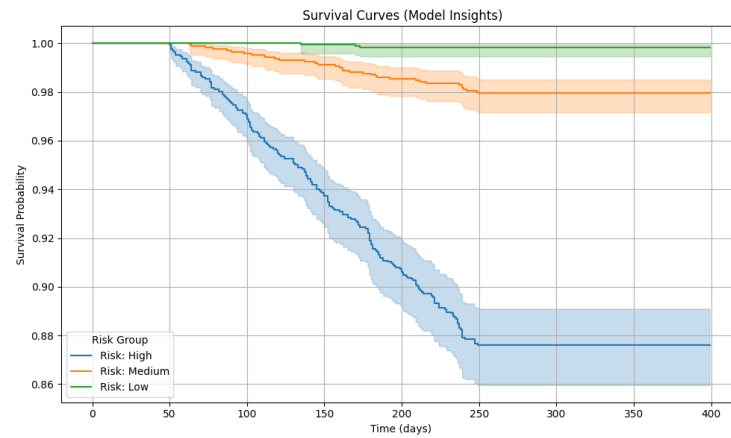


Figure 4: Kaplan–Meier survival curves stratified by stroke risk groups, showing predicted survival probabilities over time (in days). Shaded areas represent confidence intervals, illustrating distinct survival trends among risk levels.

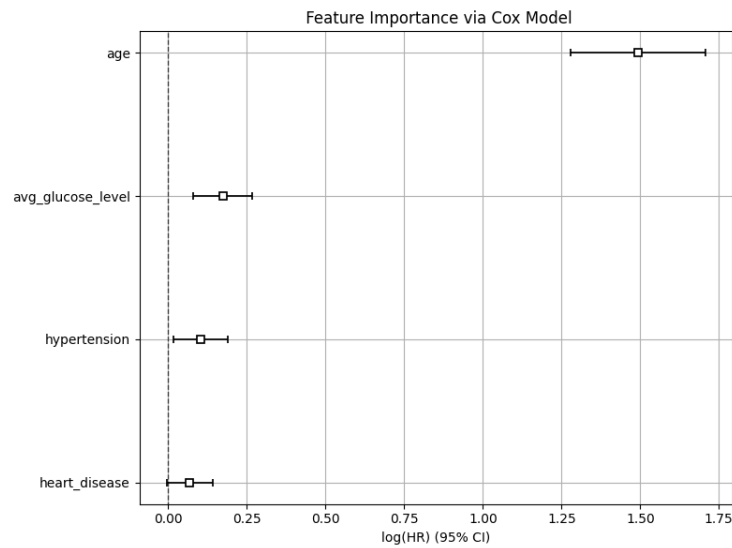


Figure 5: Cox model feature significance illustrating log hazard ratios with 95% confidence intervals for key stroke risk factors.

Non-Invasive System for Stroke Risk Stratification

9

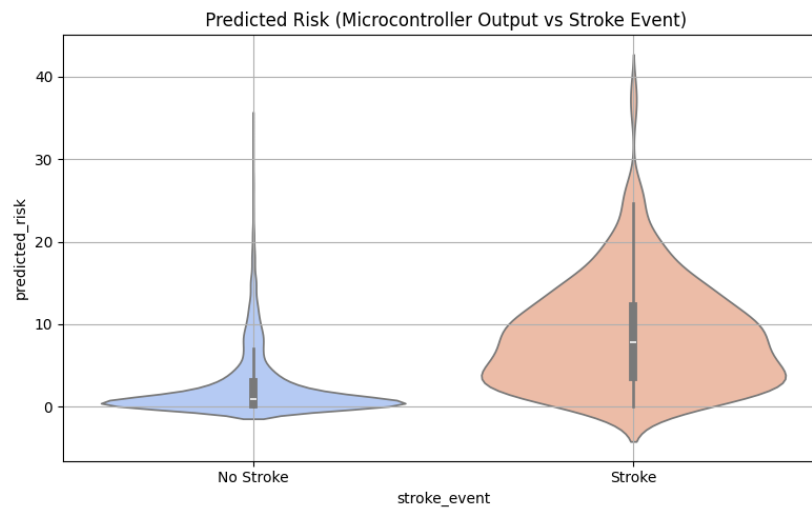


Figure6: Violin plot showing predicted stroke risk from the microcontroller across actual stroke outcomes. Higher predicted risk scores are observed for patients who experienced stroke events.

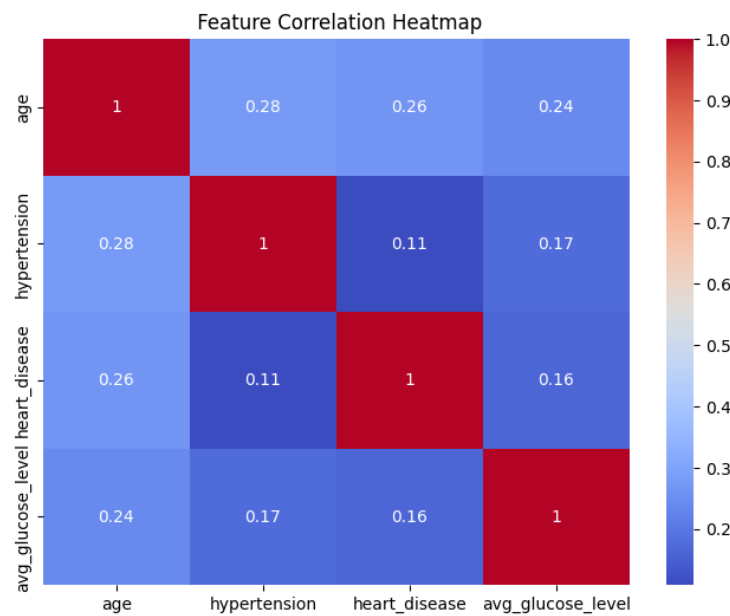


Figure 7: Heatmap showing the Pearson correlation between key cerebrovascular risk factors. Correlation intensity is color-coded, with stronger correlations indicated by warmer colors.

10 Arshi Zameer et al.

Cox Proportional Hazards model (Figure 5) quantified the impact of each variable on survival. Age emerged as the most contributing factor, with a hazard ratio (HR) of approximately 1.6, indicating a strong positive association with mortality. The average glucose level also contributed moderately to risk, while hypertension and heart disease showed lower but relevant associations. These findings are consistent with clinical literature on stroke prognosis. A violin plot (Figure 6) visualized the predicted stroke risk distribution between groups. Those who had previously experienced a stroke showed much higher predicted risk scores - skewed to the right with a median of about 7, while non-stroke individuals had scores mostly under 2. This showcases the model's ability to distinguish between the two groups. The correlation heatmap (Figure 7) showed the relationships among certain features. Age comes out to be the most correlated to hypertension and heart disease, with $r=0.28$ and $r=0.26$, respectively. All these measures enhance the model's strength in predicting risk using multiple factors.

4.2 Summary of Analytical Findings

This section provides a concise overview of the core findings and their implications derived from the study's analytical components. The results are summarized in Table 3, which highlights the methodologies, key insights, and their significance.

Table 3: Consolidated Insights from Analytical Results

Analysis Method	Key Insight
Signal Simulation (ESP32)	Successfully mimicked optical signal behavior using IR LED and photodiode with ESP32 ADC
Kaplan–Meier Survival Curves	High-risk: survival drops to 87.5% (Day 250); Medium-risk: 97.6% (Day 400); Low-risk: remains near 100%
Cox Proportional Hazards Model	Age (HR ≈ 1.6) most significant; glucose level moderately influential; hypertension and heart disease less but relevant
Predicted Risk Distribution	Stroke group: median predicted risk ≈ 7 (right-skewed); No-stroke group: mostly < 2
Correlation Heatmap	Age correlates with hypertension ($r = 0.28$) and heart disease ($r = 0.26$); other features show weak-to-moderate correlations

4.3 Discussion

All the above insights proposed that a low-cost, portable, real-time stroke risk monitoring system can be made by effectively consolidating the hardware and software strength by combining ESP32-based setup for capturing biosignals and online survival analysis and risk stratification. This real-time system becomes an effective tool for low-resource settings such as remote and rural areas. The system will not only be easy but affordable for ordinary people to use.

Conclusion and Future Work

This research introduces a conceptual model for stroke detection. The findings, which highlight the significance of different features and performance metrics, demonstrate how simulation based on data can assist in validating and optimizing designs prior to their physical execution. Mimicking such conceptual systems enables cost-effective, early-stage decision-making, and risk reduction. This foundational work can be expanded into a practical system by adding multi-wavelength LEDs, real-time signal processing, and data transmission modules. Its simplicity, affordability, and relevance to low-resource settings make it a promising direction for future stroke detection solutions.

In future, a physical prototype of this predictive stroke risk assessment device can be developed for real-time patient monitoring. Clinical trials and real-world deployment will help validate its effectiveness and usability in healthcare settings.

References

1. Y. Liang, J. Chen, Y. Chen, Y. Tong, L. Li, Y. Xu, and S. Wu, "Advances in the detection of biomarkers for ischemic stroke," *Front. Neurol.*, vol. 16, 2025, Art. no. 1488726.
2. L. Yang and Z. Wang, "Applications and advances of combined fMRI-fNIRS techniques in brain functional research," *Front. Neurol.*, vol. 16, 2025, Art. no. 1542075.
3. B. A. Taha, A. C. Kadhim, A. J. Addie, Q. Al-Jubouri, A. S. Azzahrani, A. J. Haider, A. N. Alkawaz, and N. Arsad, "Optical spectroscopy of cerebral blood flow for tissue interrogation in ischemic stroke diagnosis," *ACS Chem. Neurosci.*, 2025.
4. D. Ahirwar, D. Khurana, and S. R. Chowdhury, "Identification of ischemic stroke condition based on hemodynamic biomarkers," *IEEE Sens. J.*, vol. 22, no. 19, pp. 18944–18952, Oct. 2022, doi: 10.1109/JSEN.2022.3196497.
5. P. Zhu, X. Zhao, X. Chen, R. Liu, H. Ouyang, Y. Hu, B. Shi, and Y. Fan, "Detection of arterial stenosis based on synchronized signals from wearable pulse and blood flow velocity sensors," *ACS Sens.*, 2025.
6. S. Xu, X. Zeng, F. Yin, and C. Zhang, "Validating the reproducibility of a low-cost single-channel fNIRS device across hierarchical cognitive tasks," *Front. Neurosci.*, vol. 18, Apr. 2024, Art. no. 1351341, doi: 10.3389/fnins.2024.1351341.
7. G. Cai, J. Xu, C. Zhang, J. Jiang, G. Chen, J. Chen, Q. Liu, G. Xu, and Y. Lan, "Identifying biomarkers related to motor function in chronic stroke: A fNIRS and TMS study," *CNS Neurosci. Ther.*, vol. 30, no. 7, Jul. 2024, Art. no. e14889, doi: 10.1111/cns.14889.
8. A. Kumar, S. Crawford, T.-C. Le, A. Rahimpour, L. Carbonell, A. Capps, A. Walter, D. Liu, R. Sullivan, E. Jansen, H. Hosseini, and A. Bowden, "NIRduino: A modular, Bluetooth-enabled, Android-configurable fNIRS system with dual-intensity mode built on Arduino," *bioRxiv*, 2024, doi: 10.1101/2024.12.20.24318425.
9. J. Liang, "Developing a wearable system with fNIRS & EEG multimodality classification engine & miniaturized device," in *Proc. IEEE ICSECE*, Jinzhou, China, 2024, pp. 885–891, doi: 10.1109/ICSECE61636.2024.10729318.
10. Z. Yang, L. Ye, L. Yang, Q. Lu, A. Yu, and D. Bai, "Early screening of post-stroke fall risk: A simultaneous multimodal fNIRS-EMG study," *CNS Neurosci. Ther.*, vol. 30, no. 9, Sep. 2024, Art. no. e70041, doi: 10.1111/cns.70041.

12 Arshi Zameer et al.

11. K. Paulmurugan, V. Vijayaragavan, S. Ghosh, P. Padmanabhan, and B. Gulyás, "Brain-computer interfacing using functional near-infrared spectroscopy (fNIRS)," *Biosensors*, vol. 11, no. 10, p. 389, 2021.
12. Q. Yang and S. Ge, "Classification of a single channel fNIRS signal for a brain computer interface," in *Proc. IEEE BMEI*, Dalian, China, 2014, pp. 46–50, doi: 10.1109/BMEI.2014.7002740.
13. D. M. V. Sato, L. K. Mantovani, J. Safanelli, V. Guessier, V. Nagel, C. H. C. Moro, N. L. Cabral, E. E. Scalabrin, C. Moro, and E. A. P. Santos, "Ischemic stroke: Process perspective, clinical and profile characteristics, and external factors," *J. Biomed. Inform.*, vol. 111, 2020, Art. no. 103582.
14. W. L. Chen, J. Wagner, N. Heugel, J. Sugar, Y. W. Lee, L. Conant, M. Malloy, J. Heffernan, B. Quirk, A. Zinos, S. A. Beardsley, R. Prost, and H. T. Whelan, "Functional near-infrared spectroscopy and its clinical application in the field of neuroscience: Advances and future directions," *Front. Neurosci.*, vol. 14, p. 724, Jul. 2020, doi: 10.3389/fnins.2020.00724.
15. Fedesoriano, "Stroke prediction dataset," Kaggle, 2021. [Online]. Available: <https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset>
16. M. K. Goel, P. Khanna, and J. Kishore, "Understanding survival analysis: Kaplan-Meier estimate," *Int. J. Ayurveda Res.*, vol. 1, no. 4, pp. 274–278, Oct. 2010, doi: 10.4103/0974-7788.76794.